

All Parametric Survival Models

METIMMOX-1 Economic Evaluation

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1 Overview

This technical document presents the regression coefficients for **all parametric survival models** fitted in the METIMMOX-1 economic evaluation. While the primary analysis uses the best-fitting model (selected by AIC), this document provides full transparency by showing coefficients for all distributions tested.

1.1 Model Specification

All models use the following covariate structure (full model with age and sex adjustment):

Overall Survival (OS):

`Surv(OSwk, Death) ~ Age + sex + Rx + crp:Rx + tlr:Rx + tmb_braf:Rx`

Progression-Free Survival (PFS):

`Surv(PFSwk, Progression) ~ Age + sex + Rx + crp:Rx + tlr:Rx + tmb_braf:Rx`

1.2 Distributions Tested

Nine parametric distributions were fitted to each outcome:

1. **Exponential** - Constant hazard rate
2. **Weibull (AFT)** - Accelerated failure time parameterization
3. **Weibull (PH)** - Proportional hazards parameterization
4. **Log-logistic** - Allows for non-monotonic hazard
5. **Log-normal** - Log-transformed normal distribution for survival times
6. **Gamma** - Two-parameter distribution, flexible shape
7. **Gompertz** - Hazard increases exponentially with time
8. **Generalized Gamma** - Three-parameter family including Weibull, gamma, and log-normal as special cases
9. **Generalized F** - Four-parameter family, most flexible

2 Model Selection Summary

2.1 Overall Survival (OS) Models

Table 1: Overall Survival model fit comparison (sorted by AIC)

Distribution	AIC	BIC	Delta AIC	Rank	Selected
Gamma	622.56	646.47	0.00	1.0	***
Weibull (AFT)	623.44	647.36	0.88	2.5	
Weibull (PH)	623.44	647.36	0.88	2.5	
Log-logistic	623.68	647.60	1.12	4.0	
Log-normal	624.09	648.01	1.53	5.0	
Generalized Gamma	624.56	650.65	2.00	6.0	
Generalized F	626.56	654.82	4.00	7.0	
Gompertz	633.09	657.01	10.53	8.0	
Exponential	654.14	675.88	31.58	9.0	

2.2 Progression-Free Survival (PFS) Models

Table 2: Progression-Free Survival model fit comparison (sorted by AIC)

Distribution	AIC	BIC	Delta AIC	Rank	Selected
Gamma	453.43	477.35	0.00	1.0	***
Log-normal	453.46	477.38	0.03	2.0	
Weibull (AFT)	454.19	478.11	0.76	3.5	
Weibull (PH)	454.19	478.11	0.76	3.5	
Log-logistic	455.16	479.08	1.73	5.0	
Generalized Gamma	455.28	481.37	1.85	6.0	
Generalized F	457.28	485.55	3.85	7.0	
Gompertz	467.57	491.49	14.14	8.0	
Exponential	483.52	505.26	30.09	9.0	

3 Overall Survival (OS) Model Coefficients

Best-fitting distribution: gamma (AIC: 622.56)

The following tables present the regression coefficients for each fitted OS model. Parameters include:

- **Distribution-specific parameters** (e.g., shape, scale, rate)
- **Age** - Effect per year of age
- **Sex** - Effect of male vs female
- **Rx** - Main treatment effect (experimental vs control)
- **Biomarker:Rx interactions** - Treatment effect modification by biomarker status

3.1 Gamma (Best Fit)

Parameter	Estimate	SE	exp(Est)
shape	1.2275	0.1820	3.4126
rate	-2.8975	0.6157	0.0552
Age (per year)	0.0015	0.0082	1.0015
Sex (male vs female)	0.1454	0.1482	1.1565
RxExperimental arm	-0.2309	0.2852	0.7939
RxControl arm:crp1	-0.3570	0.3162	0.6998
RxExperimental arm:crp1	-0.3395	0.2265	0.7122
RxControl arm:tlr1	-0.8039	0.2509	0.4476
RxExperimental arm:tlr1	-0.4109	0.2129	0.6630
RxControl arm:tmb_braf1	-0.0866	0.2210	0.9171
RxExperimental arm:tmb_braf1	-0.1859	0.2072	0.8303

3.2 Exponential

Parameter	Estimate	SE	exp(Est)
rate	-4.4248	1.1022	0.0120
Age (per year)	0.0063	0.0156	1.0064
Sex (male vs female)	0.1835	0.2830	1.2014
RxExperimental arm	-0.1916	0.5253	0.8256
RxControl arm:crp1	-0.5534	0.6427	0.5750
RxExperimental arm:crp1	-0.4162	0.4332	0.6595
RxControl arm:tlr1	-0.9110	0.4699	0.4021
RxExperimental arm:tlr1	-0.5249	0.4025	0.5916
RxControl arm:tmb_braf1	-0.1924	0.4242	0.8250
RxExperimental arm:tmb_braf1	-0.2974	0.3956	0.7428

3.3 Generalized F

Parameter	Estimate	SE	exp(Est)
mu	4.1162	0.6324	61.3275
sigma	-0.6103	0.1368	0.5432
Q	0.5267	0.4284	1.6933
P	-6.7289	52.3063	0.0012
Age (per year)	-0.0014	0.0082	0.9986
Sex (male vs female)	-0.1449	0.1492	0.8651
RxExperimental arm	0.2317	0.2875	1.2607
RxControl arm:crp1	0.3559	0.3181	1.4275
RxExperimental arm:crp1	0.3405	0.2290	1.4056
RxControl arm:tlr1	0.8043	0.2520	2.2351
RxExperimental arm:tlr1	0.4124	0.2174	1.5104
RxControl arm:tmb_braf1	0.0878	0.2243	1.0918
RxExperimental arm:tmb_braf1	0.1875	0.2126	1.2062

3.4 Generalized Gamma

Parameter	Estimate	SE	exp(Est)
mu	4.1164	0.6317	61.3357
sigma	-0.6105	0.1365	0.5431
Q	0.5268	0.4281	1.6935
Age (per year)	-0.0014	0.0082	0.9986
Sex (male vs female)	-0.1443	0.1491	0.8656
RxExperimental arm	0.2321	0.2874	1.2613
RxControl arm:crp1	0.3559	0.3177	1.4275
RxExperimental arm:crp1	0.3407	0.2290	1.4060
RxControl arm:tlr1	0.8040	0.2517	2.2345
RxExperimental arm:tlr1	0.4126	0.2174	1.5108
RxControl arm:tmb_braf1	0.0878	0.2242	1.0918
RxExperimental arm:tmb_braf1	0.1877	0.2126	1.2065

3.5 Gompertz

Parameter	Estimate	SE	exp(Est)
shape	0.0137	0.0026	1.0138
rate	-5.5282	1.1991	0.0040
Age (per year)	0.0126	0.0170	1.0127
Sex (male vs female)	0.3678	0.2983	1.4446
RxExperimental arm	-0.3429	0.5268	0.7097
RxControl arm:crp1	-1.1290	0.6666	0.3234
RxExperimental arm:crp1	-0.6542	0.4688	0.5198
RxControl arm:tlr1	-1.4740	0.4923	0.2290
RxExperimental arm:tlr1	-0.7720	0.4253	0.4621
RxControl arm:tmb_braf1	-0.1956	0.4296	0.8224
RxExperimental arm:tmb_braf1	-0.2776	0.4175	0.7576

3.6 Log-logistic

Parameter	Estimate	SE	exp(Est)
shape	1.0880	0.1125	2.9683
scale	3.6647	0.6313	39.0442
Age (per year)	0.0028	0.0084	1.0028
Sex (male vs female)	-0.1517	0.1501	0.8592
RxExperimental arm	0.2843	0.3371	1.3288
RxControl arm:crp1	0.2045	0.3493	1.2269
RxExperimental arm:crp1	0.3428	0.2090	1.4089
RxControl arm:tlr1	0.8812	0.2840	2.4137
RxExperimental arm:tlr1	0.4299	0.2078	1.5371
RxControl arm:tmb_braf1	0.0838	0.2482	1.0874
RxExperimental arm:tmb_braf1	0.2259	0.1969	1.2535

3.7 Log-normal

Parameter	Estimate	SE	exp(Est)
meanlog	3.9003	0.6356	49.4154
sdlog	-0.5154	0.0969	0.5973
Age (per year)	-0.0013	0.0083	0.9987
Sex (male vs female)	-0.1283	0.1548	0.8796
RxExperimental arm	0.2676	0.3244	1.3068
RxControl arm:crp1	0.3716	0.3345	1.4501
RxExperimental arm:crp1	0.3860	0.2347	1.4711
RxControl arm:tlr1	0.7722	0.2727	2.1645
RxExperimental arm:tlr1	0.4592	0.2310	1.5829
RxControl arm:tmb_braf1	0.1527	0.2373	1.1650
RxExperimental arm:tmb_braf1	0.2332	0.2130	1.2627

3.8 Weibull (AFT)

Parameter	Estimate	SE	exp(Est)
shape	0.7281	0.1090	2.0711
scale	4.5338	0.5431	93.1093
Age (per year)	-0.0051	0.0081	0.9949
Sex (male vs female)	-0.1654	0.1429	0.8475
RxExperimental arm	0.2082	0.2536	1.2315
RxControl arm:crp1	0.4164	0.3091	1.5164
RxExperimental arm:crp1	0.3174	0.2229	1.3736
RxControl arm:tlr1	0.7730	0.2295	2.1663
RxExperimental arm:tlr1	0.3753	0.2012	1.4554
RxControl arm:tmb_braf1	0.0673	0.2081	1.0696
RxExperimental arm:tmb_braf1	0.1251	0.2032	1.1333

3.9 Weibull (PH)

Parameter	Estimate	SE	exp(Est)
shape	0.7281	0.1090	2.0711
scale	-9.3901	1.5795	0.0001
Age (per year)	0.0105	0.0168	1.0106
Sex (male vs female)	0.3426	0.2988	1.4086
RxExperimental arm	-0.4313	0.5275	0.6497
RxControl arm:crp1	-0.8624	0.6443	0.4222
RxExperimental arm:crp1	-0.6574	0.4639	0.5182
RxControl arm:tlr1	-1.6010	0.4921	0.2017
RxExperimental arm:tlr1	-0.7773	0.4178	0.4596
RxControl arm:tmb_braf1	-0.1394	0.4303	0.8699
RxExperimental arm:tmb_braf1	-0.2591	0.4186	0.7717

4 Progression-Free Survival (PFS) Model Coefficients

Best-fitting distribution: gamma (AIC: 453.43)

The following tables present the regression coefficients for each fitted PFS model.

4.1 Gamma (Best Fit)

Parameter	Estimate	SE	exp(Est)
shape	1.2426	0.1901	3.4647
rate	-1.8647	0.6321	0.1549
Age (per year)	-0.0009	0.0084	0.9991
Sex (male vs female)	0.1321	0.1653	1.1413
RxExperimental arm	0.4218	0.3073	1.5247
RxControl arm:crp1	0.0654	0.4253	1.0676
RxExperimental arm:crp1	-0.5754	0.2412	0.5625
RxControl arm:tlr1	-0.9764	0.2718	0.3767
RxExperimental arm:tlr1	-1.1914	0.2133	0.3038
RxControl arm:tmb_braf1	-0.1055	0.2537	0.8998
RxExperimental arm:tmb_braf1	-0.4891	0.2325	0.6132

4.2 Exponential

Parameter	Estimate	SE	exp(Est)
rate	-3.0223	1.1468	0.0487
Age (per year)	-0.0026	0.0161	0.9975
Sex (male vs female)	0.1749	0.3161	1.1911
RxExperimental arm	0.3956	0.5700	1.4853
RxControl arm:crp1	-0.0452	0.7953	0.9558
RxExperimental arm:crp1	-0.5694	0.4666	0.5659
RxControl arm:tlr1	-1.0805	0.5136	0.3394
RxExperimental arm:tlr1	-1.2426	0.4103	0.2886
RxControl arm:tmb_braf1	-0.2308	0.4812	0.7939
RxExperimental arm:tmb_braf1	-0.6400	0.4512	0.5273

4.3 Generalized F

Parameter	Estimate	SE	exp(Est)
mu	3.0807	0.6540	21.7727
sigma	-0.5735	0.1450	0.5635
Q	0.2876	0.6745	1.3332
P	-7.9587	48.4866	0.0003
Age (per year)	0.0001	0.0088	1.0001
Sex (male vs female)	-0.1382	0.1651	0.8709
RxExperimental arm	-0.3980	0.3284	0.6716
RxControl arm:crp1	0.0176	0.5004	1.0177
RxExperimental arm:crp1	0.5915	0.2421	1.8067
RxControl arm:tlr1	0.9720	0.2843	2.6432
RxExperimental arm:tlr1	1.2242	0.2382	3.4014
RxControl arm:tmb_braf1	0.0876	0.2656	1.0916
RxExperimental arm:tmb_braf1	0.4531	0.2403	1.5732

4.4 Generalized Gamma

Parameter	Estimate	SE	exp(Est)
mu	3.0858	0.6538	21.8843
sigma	-0.5736	0.1449	0.5635
Q	0.2849	0.6766	1.3296
Age (per year)	0.0000	0.0089	1.0000
Sex (male vs female)	-0.1381	0.1650	0.8711
RxExperimental arm	-0.3995	0.3282	0.6706
RxControl arm:crp1	0.0172	0.5006	1.0173
RxExperimental arm:crp1	0.5918	0.2420	1.8072
RxControl arm:tlr1	0.9706	0.2843	2.6394
RxExperimental arm:tlr1	1.2253	0.2383	3.4050
RxControl arm:tmb_braf1	0.0870	0.2655	1.0909
RxExperimental arm:tmb_braf1	0.4529	0.2401	1.5728

4.5 Gompertz

Parameter	Estimate	SE	exp(Est)
shape	0.0223	0.0052	1.0225
rate	-3.4251	1.1475	0.0325
Age (per year)	-0.0024	0.0166	0.9977
Sex (male vs female)	0.2329	0.3290	1.2623
RxExperimental arm	0.6939	0.5688	2.0015
RxControl arm:crp1	0.3533	0.8147	1.4237
RxExperimental arm:crp1	-0.9614	0.5783	0.3824
RxControl arm:tlr1	-1.6909	0.5483	0.1843
RxExperimental arm:tlr1	-1.9801	0.4837	0.1381
RxControl arm:tmb_braf1	-0.2949	0.4879	0.7446
RxExperimental arm:tmb_braf1	-1.3256	0.5974	0.2656

4.6 Log-logistic

Parameter	Estimate	SE	exp(Est)
shape	1.0833	0.1187	2.9545
scale	3.2498	0.7145	25.7854
Age (per year)	-0.0034	0.0090	0.9966
Sex (male vs female)	-0.1807	0.1679	0.8347
RxExperimental arm	-0.4030	0.3671	0.6683
RxControl arm:crp1	0.0133	0.4021	1.0134
RxExperimental arm:crp1	0.5903	0.2317	1.8045
RxControl arm:tlr1	1.0309	0.3232	2.8036
RxExperimental arm:tlr1	1.2757	0.2218	3.5812
RxControl arm:tmb_braf1	0.0161	0.2996	1.0163
RxExperimental arm:tmb_braf1	0.4208	0.2167	1.5232

4.7 Log-normal

Parameter	Estimate	SE	exp(Est)
meanlog	3.0492	0.6934	21.0982
sdlog	-0.5406	0.1029	0.5824
Age (per year)	-0.0008	0.0089	0.9992
Sex (male vs female)	-0.1413	0.1636	0.8683
RxExperimental arm	-0.3740	0.3369	0.6880
RxControl arm:crp1	0.1227	0.4559	1.1305
RxExperimental arm:crp1	0.6017	0.2410	1.8252
RxControl arm:tlr1	0.9463	0.2895	2.5761
RxExperimental arm:tlr1	1.2551	0.2341	3.5082
RxControl arm:tmb_braf1	0.0689	0.2661	1.0713
RxExperimental arm:tmb_braf1	0.4194	0.2157	1.5211

4.8 Weibull (AFT)

Parameter	Estimate	SE	exp(Est)
shape	0.7437	0.1139	2.1038
scale	3.2164	0.5353	24.9377
Age (per year)	0.0015	0.0080	1.0015
Sex (male vs female)	-0.1076	0.1647	0.8980
RxExperimental arm	-0.4631	0.2751	0.6293
RxControl arm:crp1	-0.1966	0.3853	0.8215
RxExperimental arm:crp1	0.5214	0.2572	1.6844
RxControl arm:tlr1	0.9559	0.2475	2.6011
RxExperimental arm:tlr1	1.1252	0.1914	3.0807
RxControl arm:tmb_braf1	0.1323	0.2349	1.1414
RxExperimental arm:tmb_braf1	0.5870	0.2596	1.7986

4.9 Weibull (PH)

Parameter	Estimate	SE	exp(Est)
shape	0.7437	0.1139	2.1038
scale	-6.7666	1.3964	0.0012
Age (per year)	-0.0032	0.0169	0.9968
Sex (male vs female)	0.2264	0.3463	1.2541
RxExperimental arm	0.9743	0.5936	2.6492
RxControl arm:crp1	0.4137	0.8152	1.5123
RxExperimental arm:crp1	-1.0969	0.5515	0.3339
RxControl arm:tlr1	-2.0111	0.5566	0.1338
RxExperimental arm:tlr1	-2.3671	0.4623	0.0938
RxControl arm:tmb_braf1	-0.2783	0.4940	0.7571
RxExperimental arm:tmb_braf1	-1.2349	0.5657	0.2909

5 Interpretation Guide

5.1 Distribution-Specific Parameters

Different parametric distributions have different ancillary parameters:

Distribution	Ancillary Parameters
Exponential	rate
Weibull (AFT)	shape, scale
Weibull (PH)	shape, scale
Log-logistic	shape, scale
Log-normal	meanlog, sdlog
Gamma	shape, rate
Gompertz	shape, rate
Generalized Gamma	mu, sigma, Q
Generalized F	mu, sigma, Q, P

5.2 Coefficient Interpretation

- **Estimate**: The coefficient on the log scale (for AFT models) or log-hazard scale (for PH models)
- **SE**: Standard error of the coefficient estimate
- **exp(Est)**: Exponentiated coefficient
 - For AFT models: time ratio (values >1 indicate longer survival)
 - For PH models: hazard ratio (values <1 indicate lower hazard/better survival)

5.3 Biomarker Interaction Terms

The biomarker:Rx interaction terms indicate treatment effect modification:

- **crp1:RxExperimental** - Effect of experimental treatment in CRP-positive patients (relative to CRP-negative)
- **tlr1:RxExperimental** - Effect of experimental treatment in TLR-positive patients
- **tmb_braf1:RxExperimental** - Effect of experimental treatment in TMB_BRAF-positive patients

A significant interaction suggests the biomarker predicts differential treatment benefit.

6 Summary

This document presents the full set of parametric survival model coefficients for the METIMMOX-1 economic evaluation. The best-fitting models (selected by AIC) are:

Outcome	Best Distribution	AIC
Overall Survival (OS)	Gamma	622.56
Progression-Free Survival (PFS)	Gamma	453.43

All model coefficients are provided for transparency and to support sensitivity analyses exploring alternative distributional assumptions.

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